

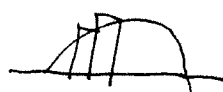
Area Between Curves

Recall Area under graph of $f(x)$ was $\int_a^b f(x) dx$

The Riemann Integral is limit of Riemann Sum

$$\int_a^b f(x) dx = \lim_{\Delta x \rightarrow 0} \sum_{k=1}^n f(x_k^*) \Delta x_k$$

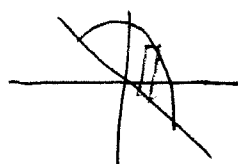
Eg Find area between $y = 2 - x^2$ and $y = x$ between $x = -\sqrt{2}$ and $x = \sqrt{2}$.



$$\sum_{k=1}^n f(x_k^*) \Delta x_k$$

$$\int_{-\sqrt{2}}^{\sqrt{2}} (2 - x^2 - x) dx = \left[2x - \frac{x^3}{3} - \frac{x^2}{2} \right]_{-\sqrt{2}}^{\sqrt{2}} \approx 3.7712$$

Eg Find the area enclosed by $y = 2 - x^2$ and $y = -x$.



- ① Draw rectangles. width = Δx
height = top - bottom = $(2 - x^2) - (-x)$

The Riemann sum becomes $\int (2 - x^2) - (-x) dx$

- ② Find limits by graphing or solving $2 - x^2 = -x$
 $x^2 - x - 2 = 0$ $(x-2)(x+1) = 0$ $x = -1, 2$

$$\int_{-1}^2 (2 - x^2) - (-x) dx = \frac{9}{2}$$

Eg Find the area between $f(x) = \sin x$, $y=0$, $x=0$, and $x=2\pi$.



Between $0, \pi$, the height is $\sin x$, but
between $\pi, 2\pi$, the height is $0 - \sin x$

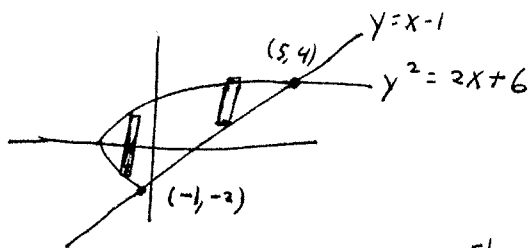
$$\sum_{k=1}^n (\sin x_k^* - 0) \Delta x_k + \sum_{k=1}^n (0 - \sin x_k^*) \Delta x_k$$

$$\boxed{\int_0^{\pi} \sin x \, dx + \int_{\pi}^{2\pi} -\sin x \, dx} = \boxed{4}$$

Set up Integral ↗

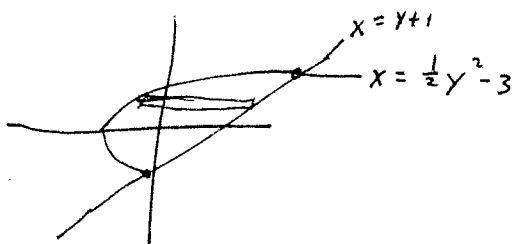
Find Area ↖

Eg Find the area enclosed by $y = x - 1$ and $y^2 = 2x + 6$



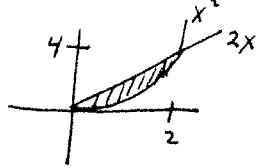
Solve
 $(x-1)^2 = (2x+6) \Rightarrow x = -1, 5$
 $y = -2, 4$

$$\int_{-3}^{-1} \sqrt{2x+6} - (-\sqrt{2x+6}) \, dx + \int_{-1}^5 \sqrt{2x+6} - (x-1) \, dx$$



$$\int_{-2}^4 (y+1) - (\tfrac{1}{2}y^2 - 3) \, dy = 18$$

Eg Find the area between $y = x^2$ and $y = 2x$.



$$\begin{aligned} x^2 &= 2x & \Rightarrow x = 0, 2 \\ \sqrt{y} &= \frac{1}{2}y & \Rightarrow y = 0, 4 \end{aligned}$$

Use x

$$\int_0^2 2x - x^2 dx = \frac{4}{3}$$

Use y

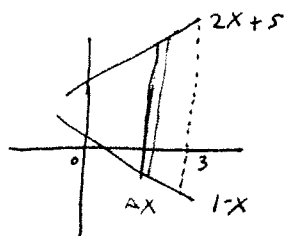
$$\int_0^4 \sqrt{y} - \frac{1}{2}y dy = \frac{4}{3}$$

Volume

Consider a solid whose base is the region bounded by

$$y = 1 - x, \quad y = 2x + 5, \quad x = 0, \quad \text{and} \quad x = 3.$$

If the cross sections are squares, set up an integral to find Volume.



Volume of box = height $\times$ width $\times$ length

This isn't a box, but we could break it into pieces, pretend they are boxes, then take limit.

$$\text{length} = (2x + 5) - (1 - x)$$

$$\text{height} = (2x + 5) - (1 - x) \quad \text{since cross sections are squares}$$

$$\text{width} = \Delta x$$

$$\text{Volume} \approx \sum_{k=1}^n \left((2x_k^* + 5) - (1 - x_k^*) \right)^2 \Delta x_k$$

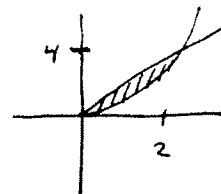
$$\text{Volume} = \int_0^3 ((2x + 5) - (1 - x))^2 dx = 237$$

Eg What if cross sections are circles? $\sum \pi \left(\frac{(2x + 5) - (1 - x)}{2} \right)^2 \Delta x$

$$\int_0^3 \pi \left(\frac{(2x + 5) - (1 - x)}{2} \right)^2 dx$$

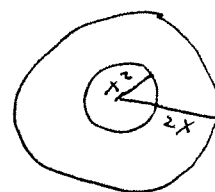
Examples of Solids of Revolution

Consider region bdd by $y = x^2$ and $y = 2x$
Find Volume when rotated about



① X-axis (Washer)

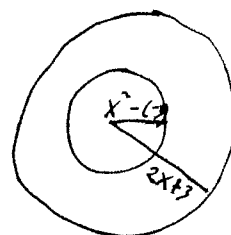
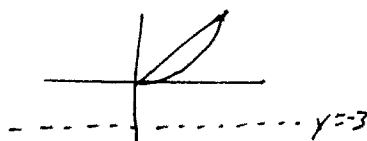
If we slice $\perp$ to x -axis, we get a washer.
Thickness = Δx



$$\sum (\pi (2x)^2 - \pi (x^2)^2) \Delta x$$

$$\int_0^2 \pi (2x)^2 - \pi (x^2) dx$$

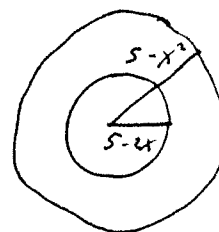
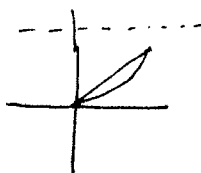
② Line $y = 3$ (Washer)



$$\sum (\pi (2x+3)^2 - \pi (x^2+3)) \Delta x$$

$$\int_0^2 \pi (2x+3)^2 - \pi (x^2+3) dx$$

③ Line $y = 5$



$$\int_0^2 \pi (5-x^2)^2 - \pi (5-2x)^2 dx$$

Important The axis is the center of the washer

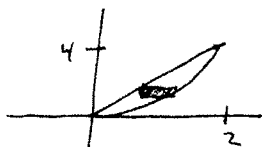
Remember that we also treated x as a function of y .

$$x = \frac{1}{2}y \quad \text{and} \quad x = \sqrt{y}$$

Slice parallel to the axis.

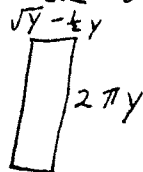
①

x -axis



We get a shell with thickness Δy .

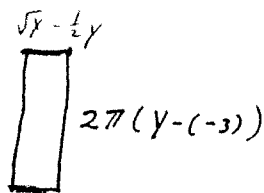
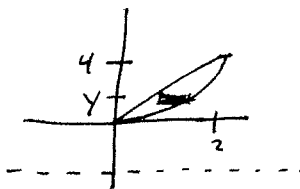
Lay out the shell



$$\sum 2\pi y (\sqrt{y} - \frac{1}{2}y) \Delta y \rightarrow \int_0^4 2\pi y (\sqrt{y} - \frac{1}{2}y) dy$$

②

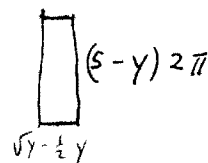
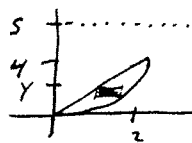
Line $y = -3$ (shells)



$$\int_0^4 2\pi (y+3) (\sqrt{y} - \frac{1}{2}y) dy$$

③

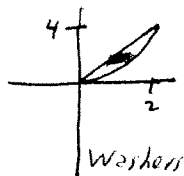
Line $y = 5$ (shells)



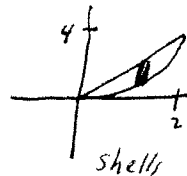
$$\int_0^4 2\pi (5-y) (\sqrt{y} - \frac{1}{2}y) dy$$

We can rotate about the y -axis as well

① y -axis

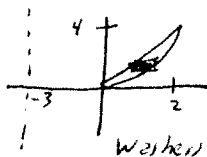


$$\int_0^4 \pi (\sqrt{y})^2 - \pi (\tfrac{1}{2}y)^2 dy$$

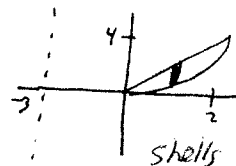


$$\int_0^2 (2x-x^2) (2\pi x) dx$$

② Line $x=-3$

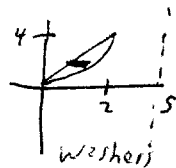


$$\int_0^4 \pi (\sqrt{y}+3)^2 - \pi (\tfrac{1}{2}y+3)^2 dy$$



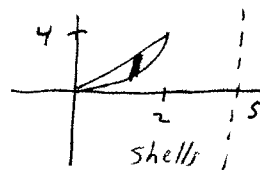
$$\int_0^2 (2x-x^2) (2\pi (x+3)) dx$$

③ Line $x=5$



~~$\int_0^4 \pi (5-\sqrt{y})^2 - \pi (5-\tfrac{1}{2}y)^2 dy$~~

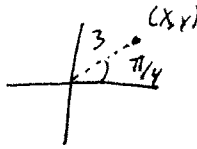
$$\int_0^4 \pi (5-\sqrt{y})^2 - \pi (5-\tfrac{1}{2}y)^2 dy$$



$$\int_0^2 (2x-x^2) (2\pi (5-x)) dx$$

Polar Coordinates

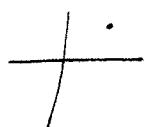
Rectangular Coordinates were horizontal/vertical directions for reaching the point.
Polar Coordinates are "as the crow flies" directions

Eg $(3, \frac{\pi}{4})$ in polar  $\cos \theta = \frac{x}{r}$, so

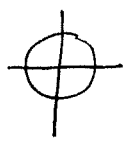
$$\begin{aligned} x &= r \cos \theta \\ y &= r \sin \theta \end{aligned}$$

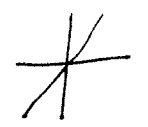
$$\begin{aligned} \sqrt{x^2 + y^2} &= r \\ \tan^{-1} \frac{y}{x} &= \theta \end{aligned}$$

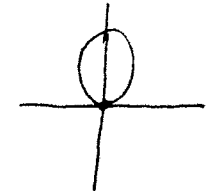
Notation $(3, \frac{\pi}{4})$ $3 \angle \frac{\pi}{4}$ $3 \text{ cis } \frac{\pi}{4}$ $3 e^{i \pi/4}$
 (r, θ) $r \angle \theta$ $r \text{ cis } \theta$ $r e^{i \theta}$

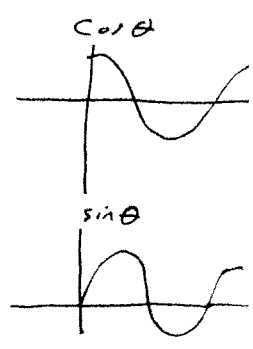
Warnings ① $(3, \frac{\pi}{4})$ and $(-3, \frac{5\pi}{4})$ and $(3, \frac{9\pi}{4})$  same point

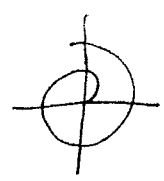
② $(0, \theta) = \text{origin } \forall \theta$

Graph $r = 3$ 

Graph $\theta = \frac{\pi}{4}$ 

Graph $r = 4 \sin \theta$ Idea is to run θ in your mind 



Graph $r = \theta$ 

Polar Area

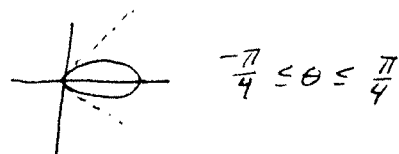
Instead of breaking the area into little rectangles, we use little sectors 
Not surprising if we think of graphing by running θ

Fact The area of sector  is $r^2 (\frac{1}{2} \Delta \theta)$

That's clear if you remember the area of a circle is $r^2 (\frac{1}{2} 2\pi)$.

Example Find the area of one leaf of $r = 4 \cos(2\theta)$

Important Always graph it first



$$\text{Riemann Sum} \quad \sum (4 \cos(2\theta_k^*))^2 (\frac{1}{2}) \Delta \theta_k$$

$$\text{Integral} \quad \int_{-\pi/4}^{\pi/4} (4 \cos 2\theta)^2 (\frac{1}{2}) d\theta$$

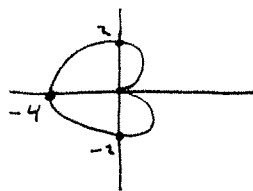
$$\int_{-\pi/4}^{\pi/4} 8 \cos^2 2\theta d\theta$$

Two Important Identities $\cos^2 x = \frac{1}{2} (1 + \cos 2x)$

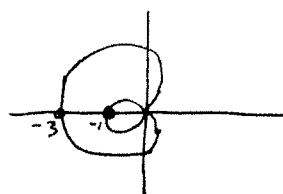
$$\sin^2 x = \frac{1}{2} (1 - \cos 2x)$$

$$\int_{-\pi/4}^{\pi/4} 8 \left(\frac{1}{2} (1 + \cos 4\theta) \right) d\theta$$

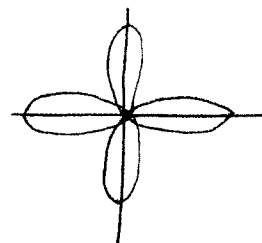
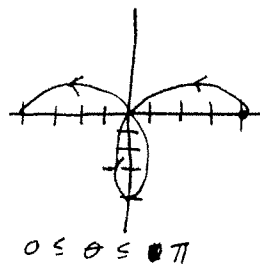
Cardioid $r = 2(1 - \cos \theta)$



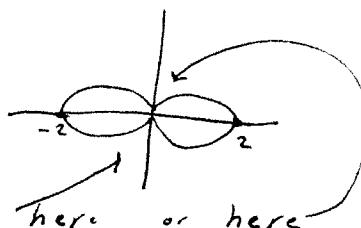
Limaçon ~~Cassini~~ $r = 1 - 2 \cos \theta$



Rose $r = 4 \cos(2\theta)$



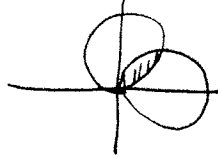
Lemniscate $r^2 = 4 \cos(2\theta)$



Warning: There is no graph here or here where $\cos(2\theta) < 0$.

Example Find the area enclosed by $r = 2 \sin \theta$ and $r = 2 \cos \theta$.

Always Graph



Find Intersections

$$2 \sin \theta = 2 \cos \theta$$

$$\tan \theta = 1$$

$$\theta = \frac{\pi}{4} \text{ or } \frac{5\pi}{4}$$

$$r = \sqrt{2} \text{ or } -\sqrt{2}$$

But that misses the intersection at origin

b/c we can only divide by $\sin \theta$ if it's not zero!

So we also have to check $\sin \theta = 0 \Rightarrow \theta = 0$.

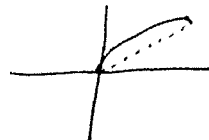
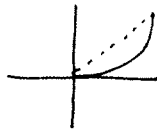
But $\theta = 0$ isn't a solution of $2 \sin \theta = 2 \cos \theta$.

Remember the origin is $(0, \theta) \forall \theta$.

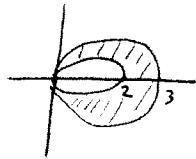
So again, Always Graph!!

$$\int_0^{\pi/4} \frac{1}{2} (2 \sin \theta)^2 d\theta$$

$$+ \int_{\pi/4}^{\pi/2} \frac{1}{2} (2 \cos \theta)^2 d\theta$$



Eg Find the area outside the lemniscate $r = 2\sqrt{\cos 2\theta}$ and inside the circle $r = 3\cos\theta$

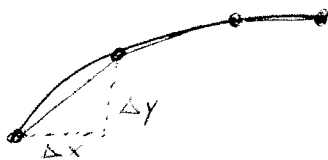


$$-\int_{-\pi/4}^{\pi/4} (2\sqrt{\cos 2\theta})^2 \frac{1}{2} d\theta + \int_{-\pi/2}^{\pi/2} (3\cos\theta)^2 \frac{1}{2} d\theta$$

Warning: This is not the same as the area between curves in rectangular coordinates.

You can't do $\int (3\cos\theta - 2\sqrt{\cos 2\theta})^2 d\theta$

Arc Length

Arc Length

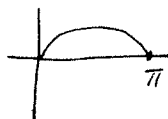
$$\Delta s^2 = \Delta x^2 + \Delta y^2$$

$$\Delta s = \sqrt{(\Delta x)^2 + (\Delta y)^2} = \sqrt{1 + \left(\frac{\Delta y}{\Delta x}\right)^2} \sqrt{(\Delta x)^2} = \sqrt{1 + \left(\frac{\Delta y}{\Delta x}\right)^2} \Delta x$$

Idea $\frac{\Delta y}{\Delta x} \rightarrow \frac{dy}{dx}$. Reason $m \vee T \Rightarrow \frac{dy}{dx} = f'(x_k^*)$

Riemann Sum $\sum \sqrt{1 + (f'(x_k^*))^2} \Delta x_k \rightarrow \int_a^b \sqrt{1 + (f'(x))^2} dx$

Eg $f(x) = \sin x$, $0 \leq x \leq \pi$



$$\int_0^\pi \sqrt{1 + (\cos x)^2} dx$$

Eg $f(x) = x^2$ on $[0, 3]$

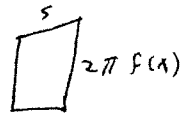
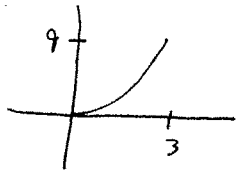
$$\int_0^3 \sqrt{1 + (2x)^2} dx$$

Eg $f(y) = \sqrt{y}$ on $[0, 9]$

$$\int_0^9 \sqrt{1 + \left(\frac{1}{2\sqrt{y}}\right)^2} dy$$

Surface Area

Rotate the graph of x^2 about the x-axis. Find surface area.



$$\int_0^3 2\pi (x^2) \sqrt{1 + (2x)^2} dx$$

Rotate about $y = -4$

$$\int_0^3 2\pi (x^2 + 4) \sqrt{1 + (2x)^2} dx$$

Rotate about $y = 15$

$$\int_0^3 2\pi (15 - x^2) \sqrt{1 + (2x)^2} dx$$

Rotate about x-axis

$$\int_0^3 2\pi x \sqrt{1 + (2x)^2} dx = \int_0^9 2\pi \sqrt{y} \sqrt{1 + \left(\frac{1}{2\sqrt{y}}\right)^2} dy$$

Polar Coord. Arc Length

$$s = \int_a^b \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

Surface Area when rotated about x-axis =

$$\int_a^b 2\pi y \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta = \int_a^b 2\pi r \sin\theta \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

Eg $r = 4 \cos(2\theta) \quad 0 \leq \theta \leq \frac{\pi}{4}$



$$\int_0^{\pi/4} \sqrt{(4 \cos 2\theta)^2 + (-8 \sin(2\theta))^2} d\theta = \text{Arc length}$$

$$\int_0^{\pi/4} 2\pi r \sin\theta \sqrt{(4 \cos 2\theta)^2 + (-8 \sin(2\theta))^2} d\theta = \text{Surface Area}$$

Work

Physics Work = Force $\times$ Distance (if force is constant)

Eg 2 lb. book moved up 5 ft $\Rightarrow$ 10 ft-lb work

Units English ft-lb Metric N·m = Joule (Aside $N = \frac{kg}{sec^2}$)

Eg Suppose the force on a particle is $F(x) = x^2$ N
How much work is done to move it from $x=0$ to $x=5$?

Split $[0, 5]$ into small intervals. Pretend $F(x)$ is constant
 $\sum F(x_k^*) \Delta x \rightarrow \int_0^5 x^2 dx = \frac{125}{3}$

Hooke's Law The force on a spring is proportional to the displacement.
 $F(x) = kx$, x = displacement.

Eg A spring is normally 5 ft long. A 4 lb force stretches it $\frac{1}{2}$ ft.

$$4 = k\left(\frac{1}{2}\right) \Rightarrow k = 8$$

How much work is done to stretch it from 6 ft to 8 ft?

$$\int_1^3 8x dx = 4x^2 \Big|_1^3 = 32 \text{ ft-lb}$$

Eg An aquarium is 1ft x 2ft x 3ft deep. Find the work done in emptying the tank. (Water weight 62.4 lb/ft³)

Slice the 3ft into pieces $\boxed{1}^2 dx$ thick.



① Let x = distance to top = distance we have to move

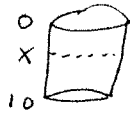
$$\int_0^3 \underbrace{x}_{\text{distance}} \underbrace{62.4}_{\text{density}} \underbrace{(1)(2) dx}_{\substack{\text{volume} \\ \text{force}}} = 561.6 \text{ ft-lb}$$

② Let x = distance from bottom $\boxed{1}^2 dx$ $3-x$ = distance we move

$$\int_0^3 (3-x) 62.4 (1)(2) dx = 561.6 \text{ ft-lb}$$

9:30 Tyler
6.1.26.24
Fanny
OK

Eg Cylindrical tank 10ft high w/ radius 6ft. Full of water. Find work to empty.

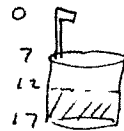


$$\int_0^{10} \underbrace{x}_{\text{distance}} \underbrace{62.4}_{\text{density}} \underbrace{(\pi 6^2)}_{\text{area}} \underbrace{dx}_{\text{depth}}$$

Force

Eg What if the tank is $\frac{1}{2}$ full and there is a 7ft pipe out top?

① Let x = distance to move



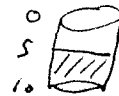
$$\int_{12}^{17} x \cdot 62.4 (\pi 6^2) dx$$

② Let x = depth



$$\int_0^5 (x+12) \cdot 62.4 (\pi 6^2) dx$$

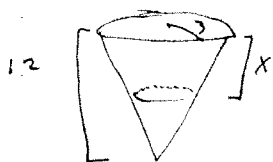
③ Let x = distance to top of tank



$$\int_5^{10} (x+7) \cdot 62.4 (\pi 6^2) dx$$

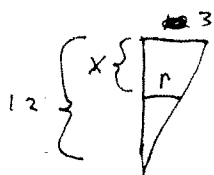
Example Conical Tank with height 12 ft, radius 3 ft.

If it is filled with water, how much work is done to pump it out?



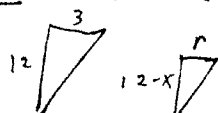
Let's let x = distance to top of tank.

$$\int_0^{12} \underbrace{x}_{\text{Distance}} \underbrace{(62.4 \text{ (Volume of slice)}) dx}_{\text{Force}}$$



$$\text{Volume} = \pi r^2 dx$$

Similar Triangles ① Same angles, so one is just a scaled version of the other



$$\frac{3}{12} = \frac{r}{12-x}$$

$$r = \frac{3}{12} (12-x)$$

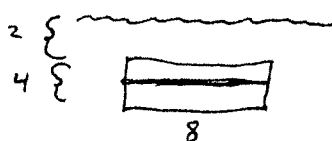
② Since right angle, $\tan \theta = \frac{3}{12} = \frac{r}{12-x}$

$$\int_0^{12} x (62.4) \pi \left(\frac{3}{12} (12-x) \right)^2 dx$$

Hydrostatic Pressure + Force

$$\text{Force} = \text{Pressure} \times \text{Area} = \text{Density} \times \text{Depth} \times \text{Area}$$

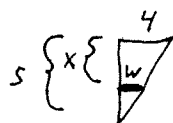
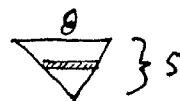
Eg A gate on a dam is a 4×8 ft rectangle. Find the force on the gate if the top is ~~2 ft under~~ ^{2 ft under} the water



① Let $x = \text{depth}$. $\int_2^6 62.4 (x) (8) dx$

② Let $x = \text{distance to top}$ $\int_0^4 62.4 (x+2) (8) dx$

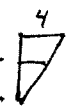
Eg A triangular gate is 3 ft under water



① Let $x = \text{distance to top}$.

$$\frac{w}{5-x} = \frac{4}{5} \Rightarrow w = \frac{4}{5} (5-x)$$

$$\int_0^5 62.4 (x+3) \left(\frac{4}{5} (5-x) \right) dx$$

② If $x = \text{depth}$, then $8-x$  $\frac{w}{8-x} = \frac{4}{5} \Rightarrow w = \frac{4}{5} (8-x)$

$$\int_3^8 62.4 (x) \left(\frac{4}{5} (8-x) \right) dx$$

Ex A semicircle level with top of water



This is the graph of $x^2 + y^2 = 5^2 \Rightarrow$

A graph of the right half of a circle centered at the origin. The horizontal axis is labeled 'x' and the vertical axis is labeled 'y'. A horizontal line segment within the circle is labeled 'w'. The radius '5' is indicated on the y-axis. The equation $x = \sqrt{25 - y^2}$ is written next to the curve.
$$x = \sqrt{25 - y^2}$$

$$W = 2x = 2\sqrt{25 - y^2}$$

$$\int_0^5 \underbrace{62.4}_{\text{density}} \underbrace{(y)}_{\text{depth}} \underbrace{(2\sqrt{25 - y^2})}_{\text{area}} dy$$

Moments + Center of Mass

Moment of a point mass about an axis = Distance to Axis $\times$ Mass

Moments measure the tendency of a body to rotate about an axis.

Recall $\exists$ 2 ways to tip down a see-saw ① More mass ② Scoot back

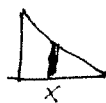
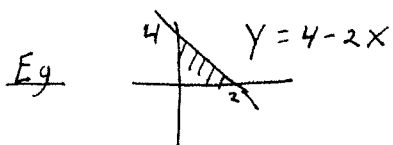
M_y = Moment about y -axis

M_x = Moment about x -axis

$$\text{Center of Mass} = (\bar{X}, \bar{Y}), \quad \bar{X} = \frac{M_y}{m}, \quad \bar{Y} = \frac{M_x}{m}.$$

(lamina)

We will assume our bodies have no thickness and density $\equiv 1$,
so mass = (1) area.



Distance to y -axis of dot = x

$$M_{\text{el}} = (4 - 2x) dx$$

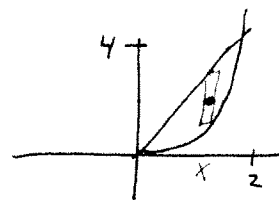
Distance to x -axis of dot = $\frac{1}{2} (4 - 2x)$

$$M_y = \int_0^2 x (4 - 2x) dx$$

$$M_x = \int_0^2 \frac{1}{2} (4 - 2x) (4 - 2x) dx$$

$$m = \int_0^2 (4 - 2x) dx$$

Eg Region bdd by $y = 2x$, $y = x^2$

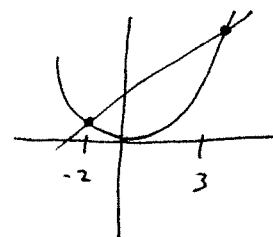


$$m_y = \int_0^2 x (2x - x^2) dx$$

$$m = \int_0^2 (2x - x^2) dx$$

$$m_x = \int_0^2 \left(\frac{2x + x^2}{2} \right) (2x - x^2) dx$$

Eg Region bdd by $y = x^2$, $y = x + 6$



$$m_y = \int_{-2}^3 x (x + 6 - x^2) dx$$

$$m_x = \int_{-2}^3 \left(\frac{x + 6 + x^2}{2} \right) (x + 6 - x^2) dx$$

$$m = \int_{-2}^3 (x + 6 - x^2) dx$$